

The MX17 Micromegas detectors under cosmic muons

Performance summary of the June 2026 Saclay cosmic-bench campaign
Reference material for the detector-construction conference talk

MX17 cosmic-bench analysis

July 2026 — preliminary

2026-07-14 note: the M3 reference recipe tightened to $\chi^2 < 1.0$, $N_{\text{clus}} = 4$ (was $\chi^2 < 5$, $N_{\text{clus}} \geq 3$) after finding the position residual was reference-limited, not chamber-limited. All numbers in this document are re-measured on the new recipe (efficiency, position resolution, and angular resolution all updated fleet-wide); the gas/drift-velocity and edge/fringe-field sections predate the change and are not expected to shift materially. See `M3_CUT_AND_ACTIVE_AREA_NOTE.md` in the analysis repository for the full investigation.

Abstract

This document summarizes, in plain language, how the five MX17 resistive-strip Micromegas detectors behave when they detect cosmic muons, based on the June 2026 test-bench campaign at Saclay. It is written as source material for a conference talk on the construction of these detectors: every headline number is quotable, every figure exists as a standalone PNG and PDF in the accompanying `figures/` and `event_displays/` directories, and a starter slide deck is provided. In one sentence: the detectors work very well — the best chamber detects $\sim 96\%$ of the muons that cross its central region, locates each crossing to about 0.47 mm (better than the 0.78 mm strip pitch), measures the track direction to about 1.7° , times the event to about 33 ns (≈ 1 mm of electron drift), and its built-in resistive spark protection works exactly as designed: discharges are rare, localized, self-quenching, and cause *no measurable dead time*.

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1 What this document is, and the one-slide summary

Five MX17 Micromegas chambers (labelled A–E at the experiment) were characterized with cosmic muons on the Saclay test bench during June 2026. The analysis behind this summary is mature: every detector was measured with the same reference telescope, the same reconstruction, and the same definitions, so the numbers below can be compared detector-to-detector and quoted directly.

Headline numbers (best detector, A, at its operating point)

Muon detection efficiency	$92.9 \pm 0.2\%$ over the active area (operating; 96.6 % spark-free); $\sim 96\%$ in t
Position resolution	$\sigma \approx 0.47\text{ mm}$ (sub-pitch; strip pitch 0.78 mm)
Track-angle resolution	$\approx 1.7^\circ$ per plane, at all track angles
Event-time resolution	33 ns ($\approx 1\text{ mm}$ of drift); 37.7 ns absolute vs. trigger
Discharge (spark) behaviour	rare (3.9 % of crossings), self-quenching, zero dead time

The five detectors are not identical in performance — two are excellent, two were operated past their sparking optimum during the tests, and one has a gas-gain problem — and Section 10 gives the honest per-detector picture with the reasons. All of the differences are consistent with operating-point and gas effects, not with construction defects.

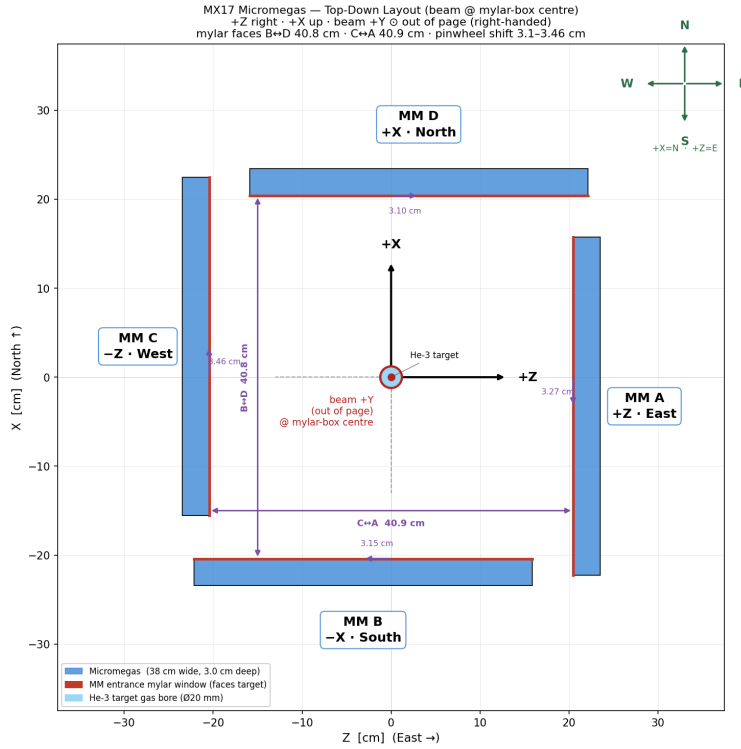


Figure 1: Where the chambers are headed: at the experiment, four of the MX17 Micromegas surround the central target in a pinwheel layout. The bench campaign described here characterized each chamber individually with cosmic muons before installation.

How to use this package. The directory layout is: `figures/` (every plot as PNG + PDF, with `FIGURE_GUIDE.md` describing each one), `event_displays/` (fresh, presentation-quality single-muon

event pictures), `slides/` (a starter PowerPoint deck), `source.reports/` (the detailed technical write-ups behind each topic), and this report.

2 The detectors and how they were tested

2.1 The device under test, as operated

Each MX17 chamber is a resistive-strip Micromegas with a two-dimensional strip readout: 512 strips in X and 512 strips in Y at a pitch of 0.78 mm, giving a $\sim 40 \times 40 \text{ cm}^2$ active area. A pixelated top layer divides the avalanche charge between the two strip layers, so a single avalanche is seen by *both* coordinates at once (Section 7 shows how evenly). Above the amplification mesh sits a **30 mm conversion-and-drift gap**: a cosmic muon crossing the chamber liberates ionization electrons along its path through this gas volume, and these drift down to the mesh where they are amplified and read out.

Operating configuration on the bench:

- **Gas:** Ar/isobutane 95/5 at atmospheric pressure, flushed continuously (the campaign also quantified what happens when moisture and air contaminate the gas — Section 8).
- **High voltage:** amplification (“resist”) voltage typically 480–490 V; drift voltage typically 1000 V across the 30 mm gap ($\approx 330 \text{ V/cm}$ drift field).
- **Readout:** DREAM front-end electronics, 512 channels per plane, sampling each strip at 60 ns intervals over a $1.92 \mu\text{s}$ window — so every strip records a full pulse *waveform*, with both an amplitude and a time.

2.2 The cosmic bench

The bench provides two independent references against which the chambers are judged:

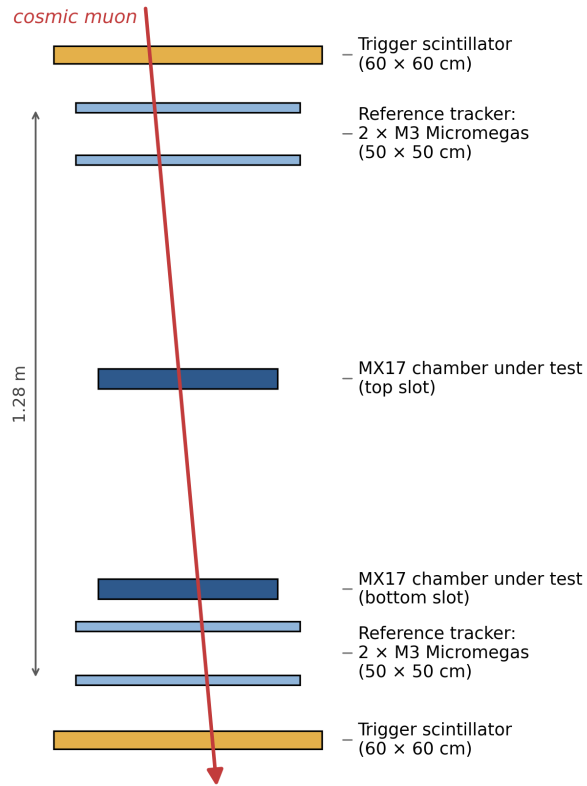
- **A reference tracker (“M3”):** a telescope of multiplexed Micromegas planes above and below the test slots. For every triggered cosmic muon it reconstructs the muon’s straight line in space, independently of the detector under test. Every performance number in this document is defined against these reference tracks: the track says where the muon truly went; the chamber under test either saw it there or it did not.
- **A scintillator trigger:** a top+bottom scintillator-paddle coincidence ($\sim 5 \text{ ns}$ precision) starts the readout, providing the absolute time reference used for the timing measurement.

Alignment of each chamber to the telescope converged to better than a millimetre in every case, and the recovered geometry (rotation $\approx 89.5^\circ$, heights consistent with the mechanical slot positions) matches the as-built bench.

2.3 The definitions used everywhere

Efficiency For every clean reference muon track, project it to the chamber plane and ask: did the chamber reconstruct a hit (with both X and Y coordinates) within 5 mm of that point? The quoted efficiency is the fraction of “yes”. A muon for which the chamber recorded nothing at all still counts in the denominator — there is no hidden normalization.

Saclay cosmic test bench (side view, to scale in height)



Readout triggered by top + bottom scintillator coincidence;
the four M3 planes reconstruct each muon independently of the chamber under test.

Figure 2: The Saclay cosmic bench (side view, heights to scale): trigger scintillators above and below, two pairs of M3 reference Micromegas, and the two MX17 test slots in between.

Position resolution The width (Gaussian core, σ) of the difference between the chamber’s reconstructed position and the reference track’s position. Note this still *includes* the small pointing error of the reference telescope itself, so all position resolutions quoted here are slightly conservative (upper limits on the true detector resolution).

Angular resolution The spread (σ or the 68% half-width, σ_{68}) of the difference between the track angle measured by the chamber alone (Section 3) and the reference track angle.

Spark fraction Discharges fire most of the detector at once (>50 strips); they are counted separately as the percentage of muon crossings that coincide with such a discharge.

3 How the chamber sees a muon: micro-TPC operation

The 30 mm drift gap is not just a conversion volume — it turns each chamber into a small time-projection chamber (“micro-TPC”). Electrons freed deeper in the gap arrive at the mesh *later*: at the nominal drift field the electrons travel at $34\text{ }\mu\text{m}/\text{ns}$, so the 30 mm gap maps onto a $\sim 0.9\text{ }\mu\text{s}$ spread of arrival times, comfortably resolved by the 60 ns sampling. Each fired strip therefore contributes a point (strip position, arrival time) = (horizontal position, depth), and a straight-line fit through those points gives the muon’s crossing angle *from a single chamber* — no external tracker needed.

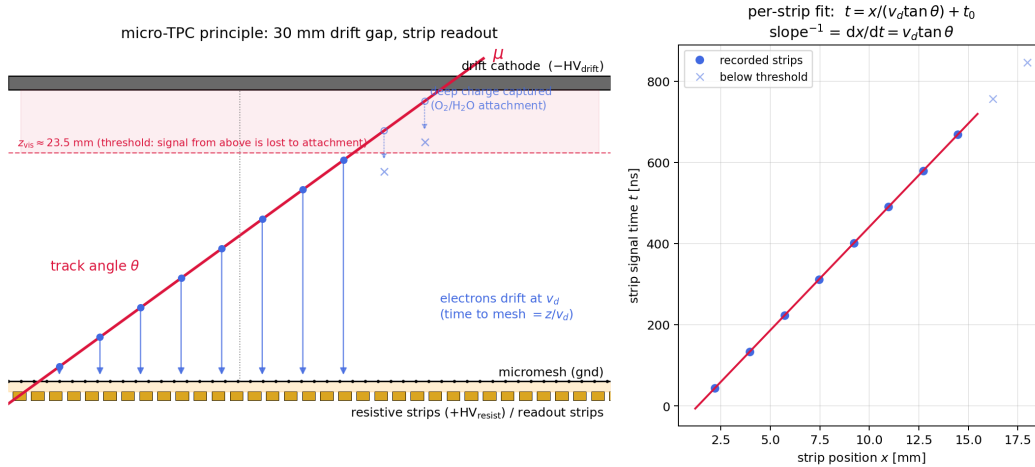


Figure 3: The micro-TPC principle: an inclined muon (left) produces a ladder of strip signals whose arrival times encode depth (right); the slope of the ladder is the track angle.

3.1 Event displays

Figures 4 and 5 are real single-muon events from detector A, drawn directly from the raw data. In the strip-vs-time view the muon is simply a straight line of charge; in the raw waveform view the diagonal charge stripe across the channel-vs-time plane is the muon track as the electronics actually record it. Figure 6 puts the two strip planes together into a single 3-D picture: the drift time of each strip signal becomes a depth, so the charge cloud traces the muon’s path through the gas — and the independent reference telescope track (green) runs straight through it, a direct visual check of both the reconstruction and the alignment.

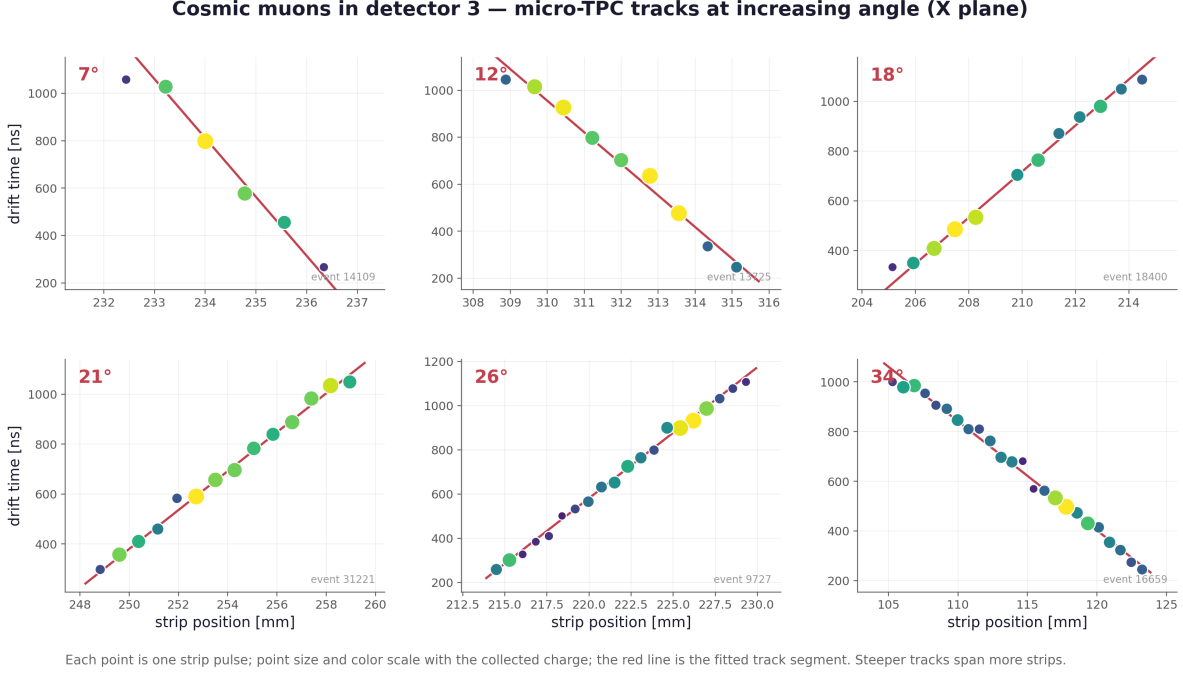


Figure 4: Gallery of cosmic-muon events in detector A. Each panel shows one muon crossing: every marker is one strip signal, placed at its strip position (horizontal) and electron drift time (vertical, equivalently depth in the gas), colored by pulse amplitude. The overlaid line is the track fitted from this chamber’s own data.

3.2 A built-in subtlety: charge sharing between strips

The resistive strips deliberately spread the avalanche charge across neighbours: a strip receives $\sim 45\text{--}52\%$ of its neighbour’s prompt charge (and a few percent at two strips), with a $\sim 70\text{ ns}$ delay, and the spread reaches $\sim 3\times$ further along the resistive-strip direction than across it. This was measured directly from vertical tracks and is the same on both chambers tested — it is a *design property*, not a defect. It cuts both ways:

- It distorts the naive time ladder (shared charge arrives with the neighbour’s timing), which biases simple angle fits; the analysis removes this with an “unsharing” correction, after which the time-based and geometry-based measurements agree.
- It is repaid in full for position: the sharing acts as a built-in charge interpolator between strips, which is precisely why the position resolution beats the strip pitch (Section 4.2).

4 Headline performance of the best chamber (Detector A)

All numbers in this section are from Detector A (mx17_3) at its operating point (490 V amplification, 1000 V drift), using the highest-statistics June runs (22,300 clean reference muons in the headline run at the current recipe — fewer than the previous 52,000 because the stricter reference-track cut trades statistics for a cleaner reference, see the note on page 1).

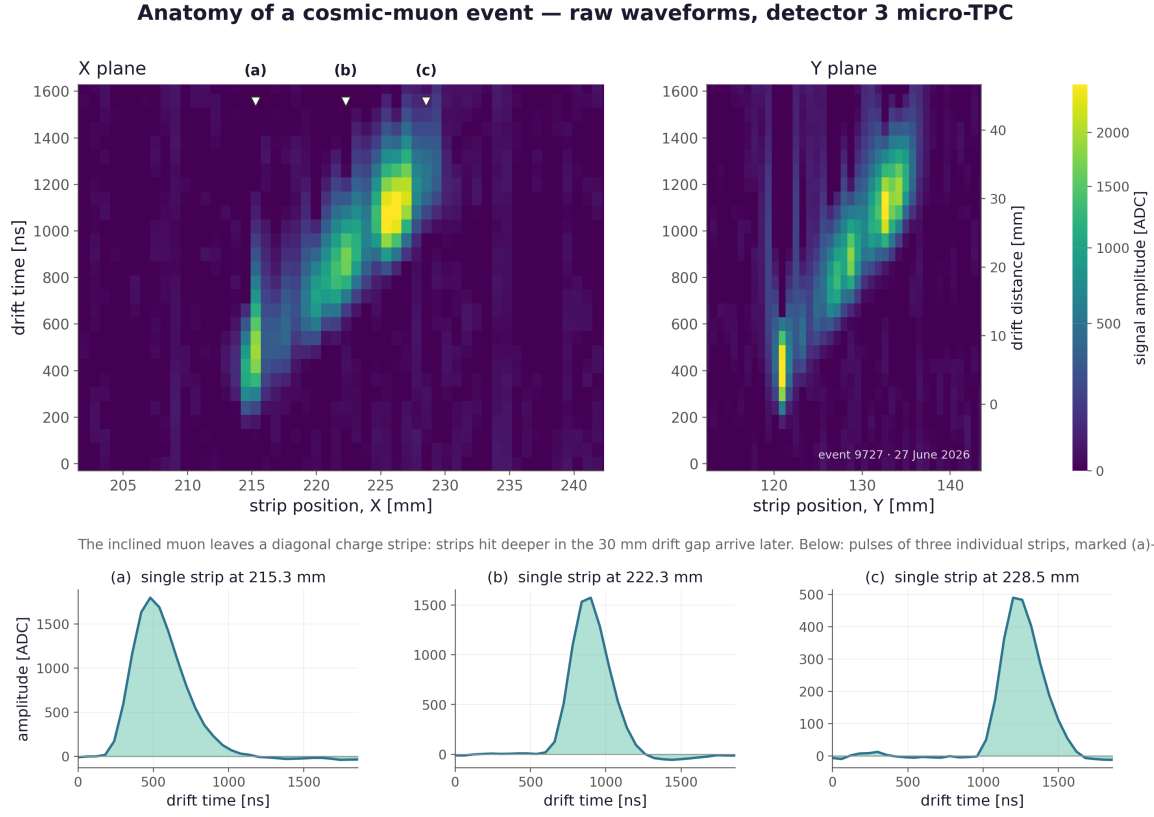
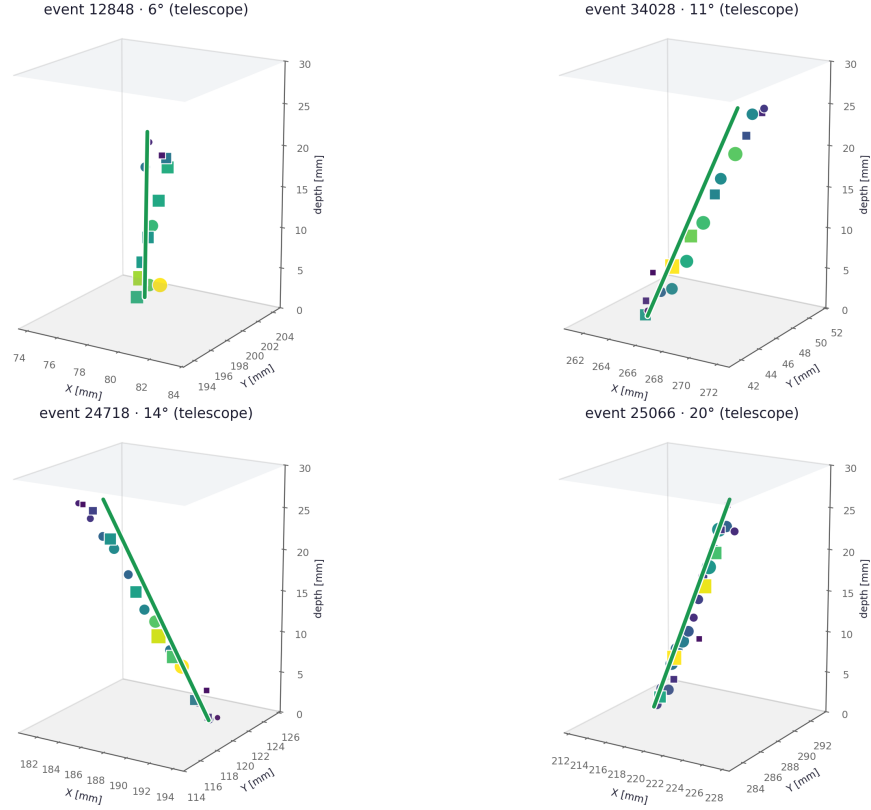


Figure 5: “Anatomy of an event”: the raw DREAM waveforms for one inclined muon. The diagonal stripe in the channel-vs-time plane is the muon; its slope is the drift-time ladder used for the angle measurement.

● X strips
 ■ Y strips
 — M3 reference track

Cosmic muons in 3-D — charge cloud vs M3 reference track

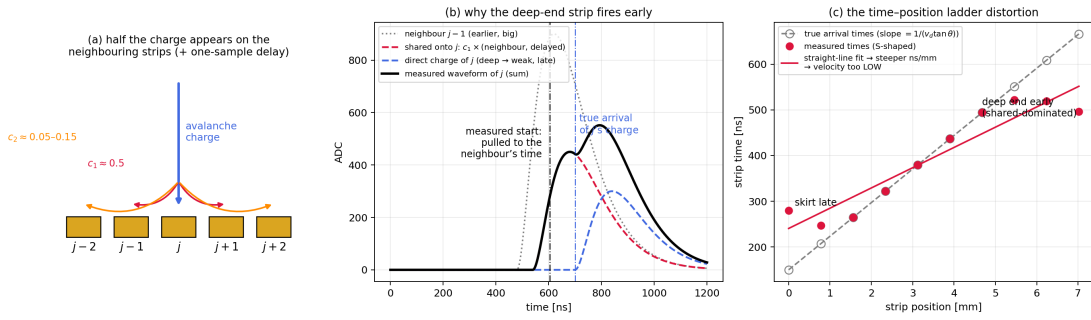


Each point is one strip signal (circle = X plane, square = Y plane) placed at its strip position and drift depth, colored by amplitude; the green line is the independent M3 telescope track.

detector A · 27 June 2026

Figure 6: Four real events in 3-D, spanning $\sim 6\text{--}20^\circ$. Each marker is one strip signal (circle = X plane, square = Y plane) placed at its strip position and at the depth implied by its electron drift time, colored by pulse amplitude. The green line is the *independent* M3 telescope track, extrapolated into the chamber — it threads the charge cloud in every event. These are among the tightest matches (0.1–0.3 mm at the mesh); across the full sample the chamber-to-reference match has a 0.6–0.8 mm per-axis core (Section 4.2), so a *typical* event sits ~ 0.8 mm off in this 2-D radial sense — the display events were chosen near the tight end to show the geometry cleanly. Any residual fanning of the raw per-strip ladder away from the line with depth is the resistive-strip charge-sharing bias (Section 3.2), removed before angles are quoted — it is not a position error. Single-event versions (with turntable animations) are in `event_displays_3d/`.

Charge sharing on resistive strips: mechanism of the time-fit velocity bias



mx17_det3_saturday_scan_6-27-26 — charge sharing measured from vertical tracks

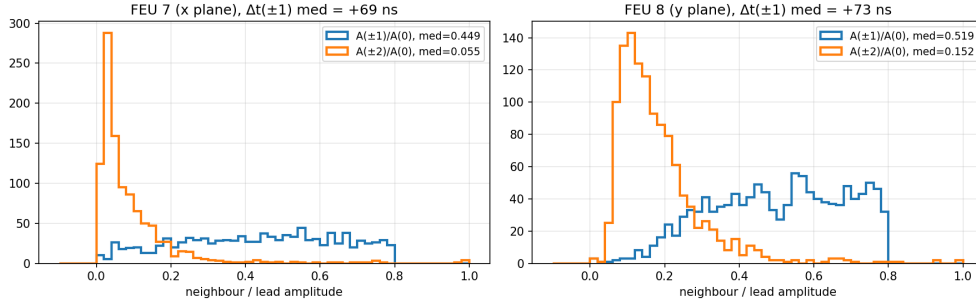


Figure 7: Top: how resistive-strip charge sharing works and why it distorts a naive time fit. Bottom: the sharing measured in data from vertical tracks — the first neighbour carries $\sim 45\%$ (X) / 52% (Y) of the lead strip's charge, with a ~ 70 ns delay.

4.1 Detection efficiency

Efficiency — Detector A

92.9 ± 0.2 % of reference muons are reconstructed within 5 mm over the full active area — the operating number, with the 3.9 % in-spark coincidence folded in (spark-free it is 96.6 ± 0.1 %); **~ 96 %** in the core (>25 mm from the frame, pre-2026-07-14 figure, not yet re-verified). Crucially, the chamber produces a signal for **100.0 %** of muons and a reconstructed track point for 96.0 %: genuine blindness is effectively 0.0 %. The loss off 92.9 % is a discharge coincidence (3.9 %) plus an edge/near-miss *position* tail (3.1 %), not dead area.

The efficiency is uniform across the chamber face except for the narrow edge band: Figure 8 shows the map. Three numbers describe the same chamber and it is worth keeping them straight: **92.9 %** is the operating efficiency (everything, including the muons that arrive during a discharge); **96.6 %** is the same measurement on spark-free crossings; **~ 96 %** is the core, away from the ~ 25 mm fringe-field edge band that is present by construction in any bounded drift volume. The 3.1 % that reconstructs *past* the 5 mm cut is a near-miss position tail (competing clusters, edge distortion, sub-veto discharges), almost all inside 10 mm — loosening the match recovers still more of it. The chamber essentially never fails to *see* a muon; the question is only how close the reconstructed point lands.

4.2 Position resolution: better than the strip pitch

Position resolution — Detector A

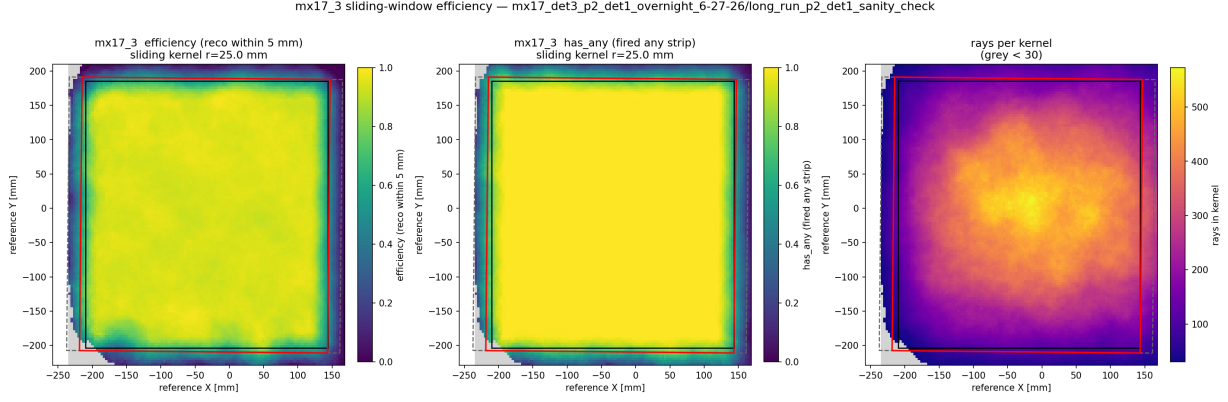
Gaussian-core residuals against the telescope: $\sigma = 0.44\text{--}0.47$ mm (radial core, $\chi^2 < 1.0 + N_{\text{clus}} = 4$ recipe), and the best position estimator reaches **0.47/0.54 mm** (X/Y) for near-vertical muons — *below the 0.78 mm strip pitch*. Sub-pitch precision is possible only because the resistive layer shares charge between neighbouring strips, providing analog interpolation. All values still include the reference telescope’s own pointing error, so the true detector resolution is slightly better.

4.3 Track-angle resolution: $\sim 1.7^\circ$ from a single chamber

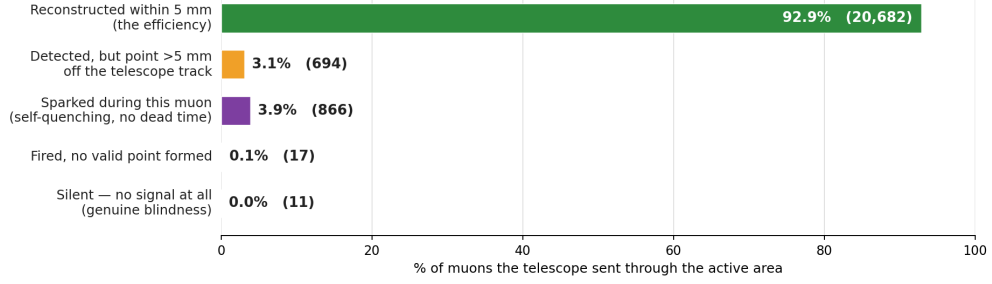
Angular resolution — Detector A (hybrid tracking)

$\sigma_{68} \approx 1.7^\circ$ per readout plane (1.66° in the near-vertical band, 1.84° on the inclined plateau), *at all track angles* including head-on, with a residual bias of $\approx 0.1^\circ$ and 97–98% of tracks usable. The same reconstruction transferred untouched to Detector B gives 2.7° (vs. 2.3° retrained), confirming this is a property of the detector design, not a tuning artifact – every chamber’s transfer behaviour matches the same pattern seen before the M3 recipe change.

Two methods are combined: for inclined tracks the drift-time ladder (micro-TPC fit, after the unsharing correction of Section 3.2); for near-vertical tracks — where the time ladder carries no lever arm — the shape of the charge deposit itself. The hybrid of the two delivers a uniform $\sim 1.7^\circ$ everywhere. In addition, the fitted drift velocity from the data agrees with the dedicated measurement to 1–3%, meaning the chambers can monitor their own gas condition in situ during the experiment.



Detector A (mx17_3): where do the crossing muons go? (22,270 muons, 5 mm match)



The chamber produces a signal for 100.0% of muons and reconstructs a track point for 96.0%. Genuine blindness (no signal at all) is only 0.0%. The two biggest "losses" off the 88.8% are NOT the chamber failing to see the muon: a 3.9% spark-coincidence (self-quenching, zero dead time after), and a 3.1% edge / near-miss position tail — almost all within 5–10 mm, so at a 10 mm match the efficiency recovers to ~95%.

Figure 8: Top: Detector A efficiency map — fraction of reference muons matched within 5 mm vs. position (left), fraction with any signal at all (middle, uniformly ~100%), and track statistics (right). Bottom: the loss budget — 92.9% reconstruct in place; the remainder is near-misses (3.1%) and in-spark coincidences (3.9%); essentially no muons leave no signal.

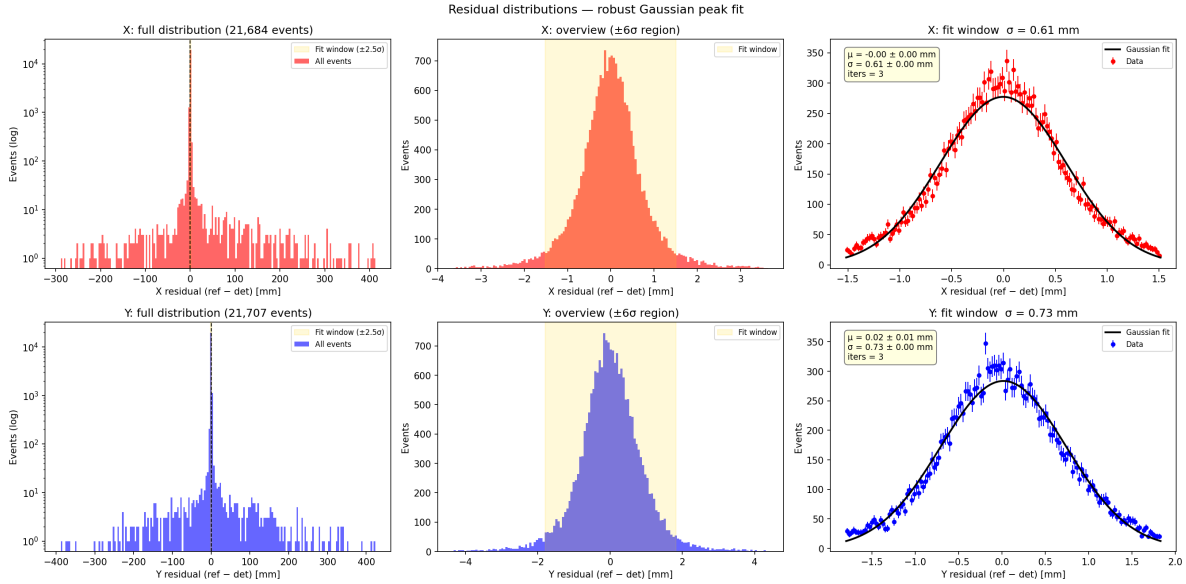


Figure 9: Detector A position residuals with respect to the reference track in X (top row) and Y (bottom row): full distribution, zoom, and Gaussian core fit ($\sigma = 0.61/0.73$ mm on this run, telescope included).

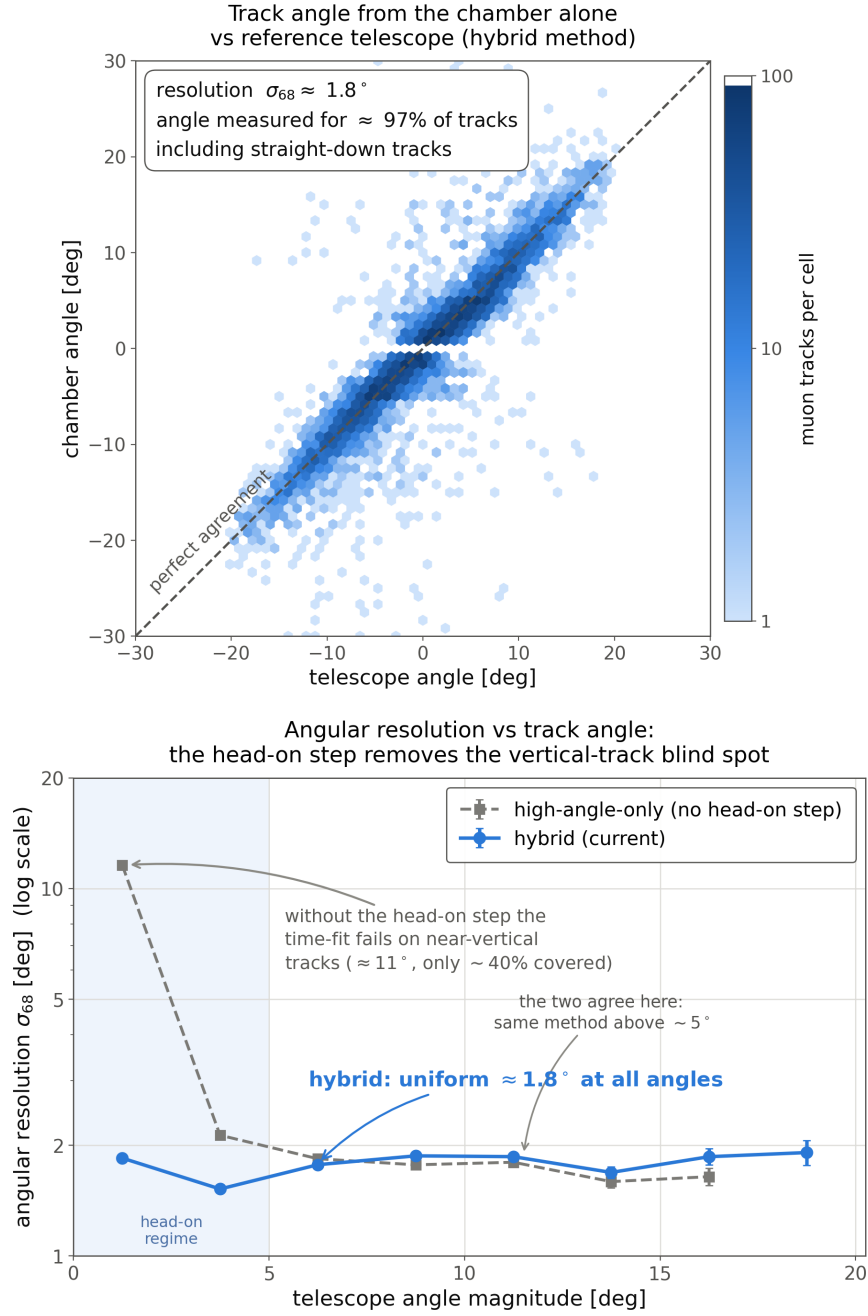


Figure 10: Hybrid tracking, Detector A. Top: track angle measured by the chamber alone vs. the reference telescope — the events hug the diagonal. Bottom: angular resolution vs. track angle. The grey curve is the *same* method with the head-on step switched off (“high-angle-only”: the drift-time fit alone); it tracks the hybrid above $\sim 5^\circ$ — they are the same estimator there — and rises to $\approx 5^\circ$ on near-vertical tracks, where a drift-time fit has no lever arm (and there covers only $\sim 51\%$ of them). The hybrid stays uniform at $\approx 1.7^\circ$ across all angles at 97–98% coverage.

4.4 Time resolution: ≈ 1 mm of drift

Time resolution — Detector A (replicated on B within 10%)

Detector time resolution $\sigma_t = 33$ ns per plane (29 ns after time-walk correction), i.e. ≈ 1 mm of electron drift — measured telescope-free from the agreement of the two orthogonal strip layers timing the same drift electrons (inter-plane bias -1.3 ns, consistent with zero). Absolute event time versus the scintillator trigger: $\sigma_{68} = 37.7$ ns, dominated by the detector term itself.

The two strip layers sit under one common amplification stage and read the same avalanche, so their time difference is a pure, assumption-free measure of the chamber's own timing — a clean consequence of the two-coordinate design. The ≈ 1 mm drift-equivalent timing matches the ≈ 0.6 mm transverse resolution: the chamber is not timing-limited relative to its geometry.

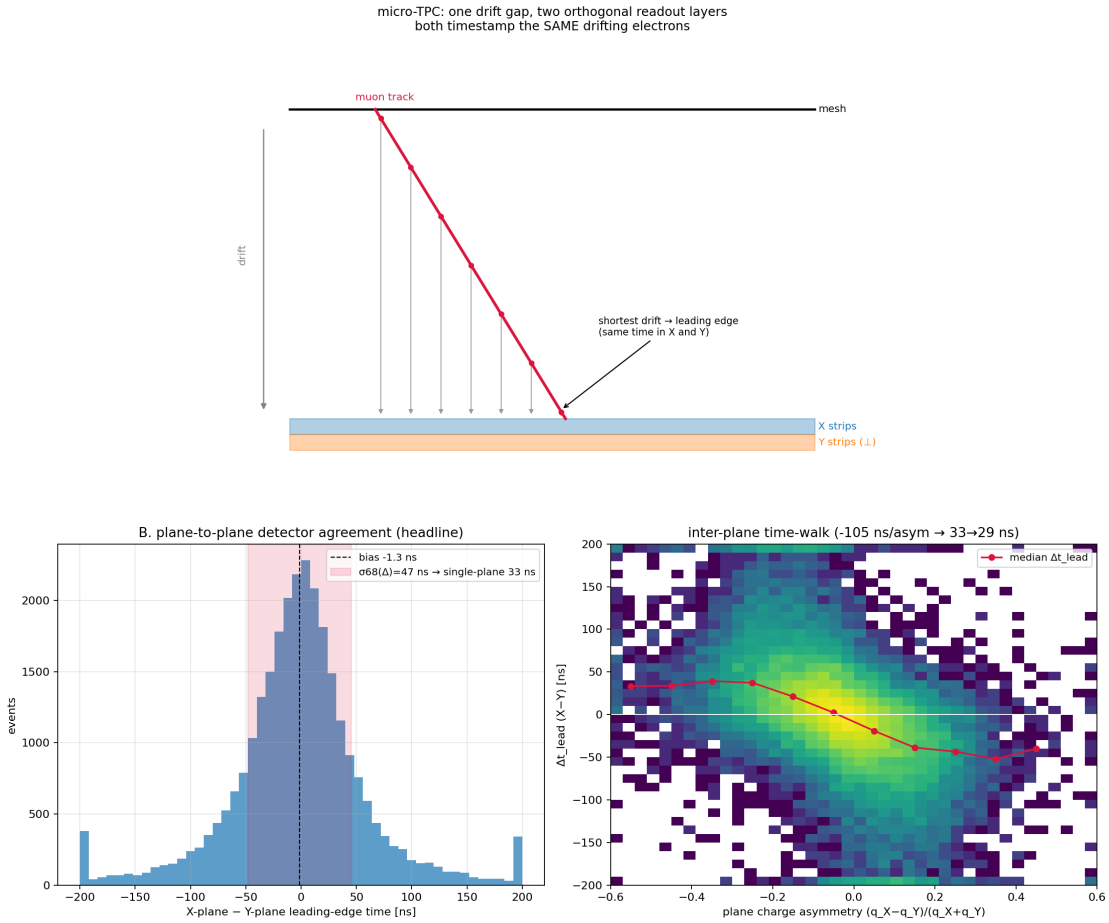


Figure 11: Top: the built-in timing measurement — the X and Y layers independently timestamp the same drifting electrons. Bottom: their time difference gives $\sigma_t = 33$ ns per plane with zero bias.

5 Operating point: high-voltage behaviour

Each detector was scanned in amplification voltage. The behaviour is textbook: efficiency turns on, reaches a plateau, then *falls* at high voltage — and the falloff is entirely explained by sparking,

not by the detector going quiet (the fraction of events with any signal stays $\sim 100\%$ throughout, while the spark fraction climbs from a few percent to $\sim 50\%$ exactly where the efficiency rolls off). Resolution is best at the efficiency optimum.

Operating points found (drift 1000 V for A/B, 700 V for C/D)			
Detector	Peak efficiency	Optimal amplification HV	Plateau behaviour
A (mx17.3)	93.2 %	490–495 V	broad plateau, $\sigma \approx 0.5\text{--}0.65$ mm
B (mx17.2)	94.9 %	485–490 V	plateau, sparks off above ~ 510 V
C (mx17.6)	78.8 %	480 V	full turn-on $\rightarrow$ plateau $\rightarrow$ falloff
D (mx17.7)	64.1 %	440 V	full turn-on $\rightarrow$ plateau $\rightarrow$ falloff

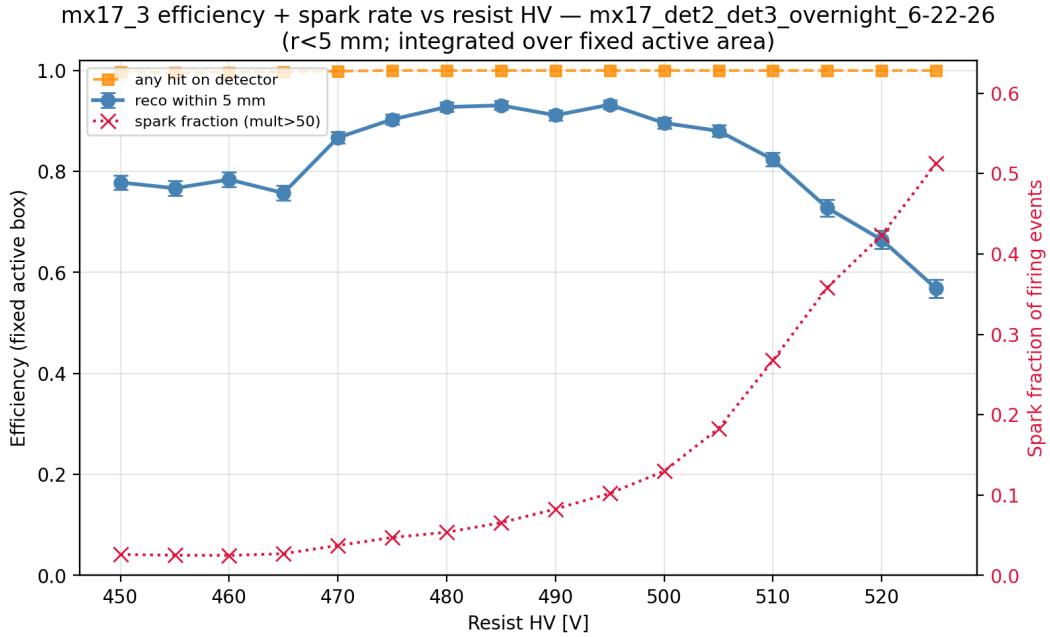


Figure 12: Detector A amplification-voltage scan: efficiency (left axis) and spark fraction (right axis). The efficiency loss above the plateau is spark-induced, not loss of signal.

6 Discharges: the resistive protection works

Because the amplification structure is resistive, a discharge (“spark”) is expected to stay small, quench itself, and leave the detector immediately live. The June campaign tested every part of that expectation directly.

Spark behaviour — measured, Detector A at operating point

Rate: 3.9% of muon crossings (0.33 Hz, Poisson — i.e. random, no bursts). Mostly muon-induced ($4\times$ enhancement in coincidence with a crossing) and concentrated at the chamber edge. Non-propagating: the raw waveforms show a fast *global baseline step* that recovers within the $1.92\,\mu\text{s}$ readout window in 94% of cases, with the genuine localized charge confined to $\sim 40\text{--}56$ strips at one edge. And critically: **no measurable dead time** — reconstruction efficiency and gas gain are flat as a function of time since the previous spark, so the only loss is the crossing that coincides with the spark itself (already inside the efficiency numbers).

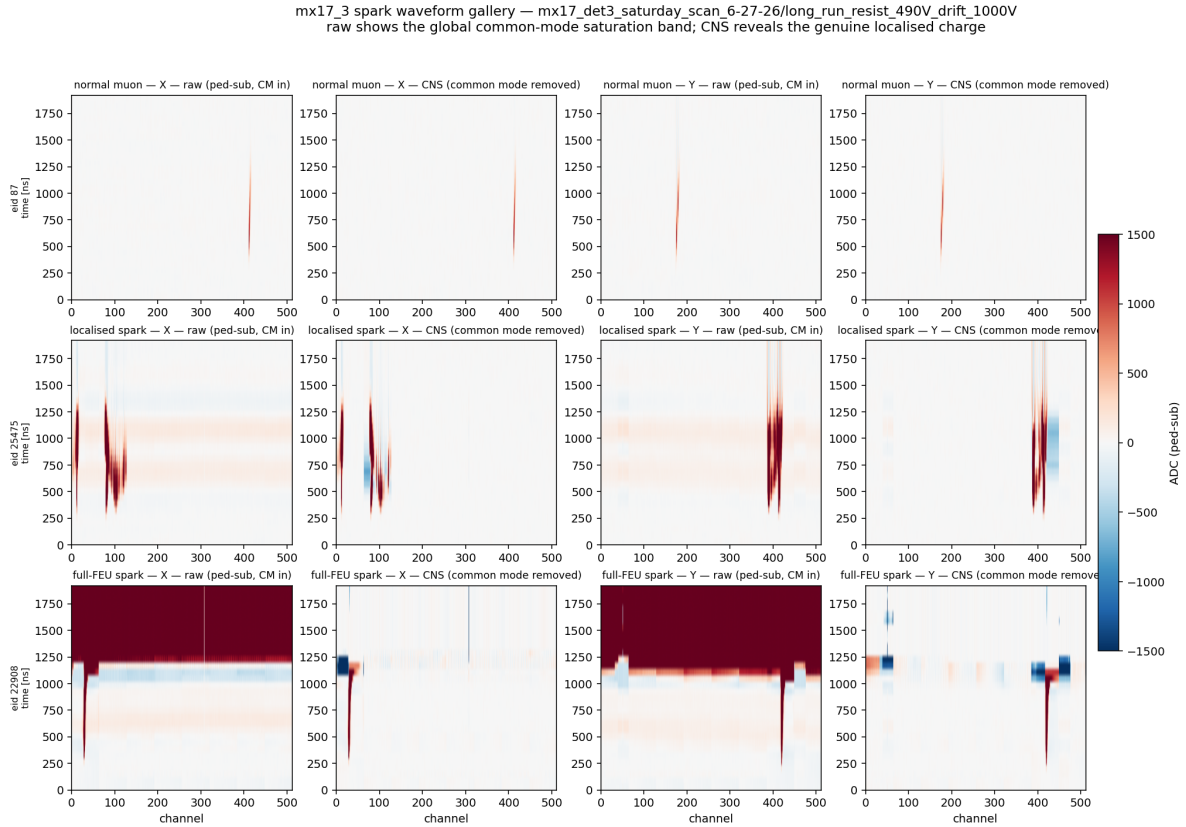


Figure 13: What a discharge actually looks like in the raw data: a fast, global, self-quenching baseline excursion (compare the clean muon of Fig. 5). It is over within the $2\,\mu\text{s}$ readout window.

For a construction talk this is arguably the strongest single message: the resistive spark-protection scheme was verified at the raw-waveform level and imposes *no recovery-time penalty* on operation.

7 The pixelated layer shares charge evenly between X and Y

The top pixelated layer is supposed to route the avalanche charge evenly to the two strip coordinates. Measured: the fraction of charge on the X layer, $f = q_X / (q_X + q_Y)$, is centred at 0.491 (A) / 0.534 (B) — within a few percent of the ideal 0.5, with the small per-chamber offset attributable to

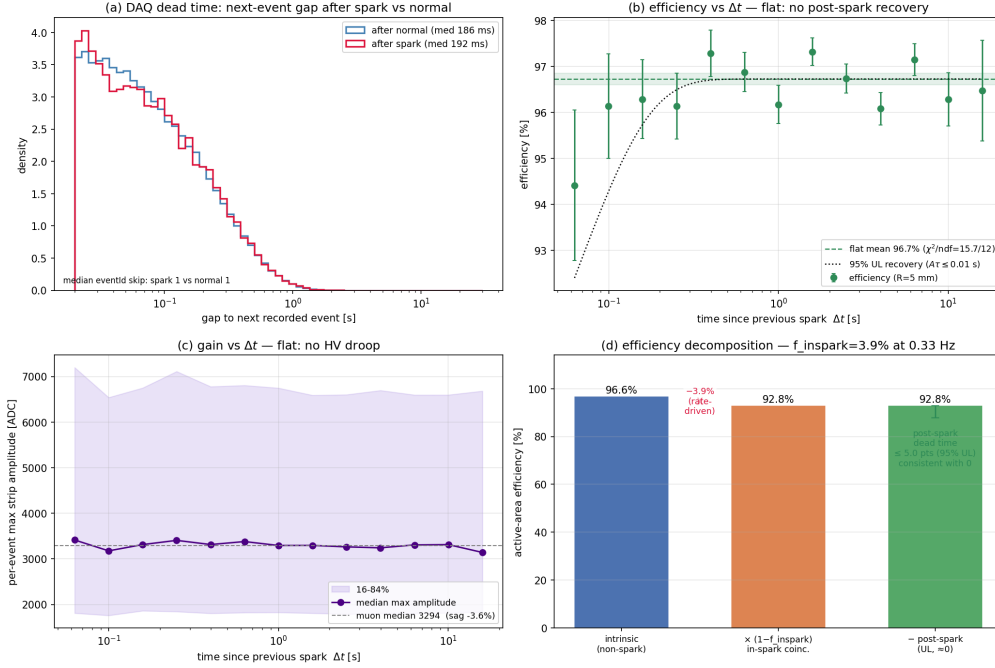


Figure 14: The dead-time null measurement: efficiency and pulse height vs. time since the previous discharge are flat — the chamber is immediately live again.

assembly — and is *constant* across the chamber face and across track angles (spread $\sigma_{68} \approx 0.07$ per event; true position-to-position variation only $\sim 2\%$). The X and Y charges of a given event are strongly correlated ($r \approx 0.84\text{--}0.89$): both layers really do see the same avalanche.

8 Gas matters: drift velocity, moisture, and attachment

The campaign measured the drift velocity at five drift fields and found $34 \pm 1.5 \mu\text{m}/\text{ns}$ at the nominal 1000 V — and the shape of the velocity-vs-field curve identifies the gas: it matches simulation (Magboltz) for Ar/iso 95/5 with $\sim 1\%$ water. Two practical lessons for operation:

- **Moisture changes the drift speed.** Detector A’s gas visibly dried from $>3\%$ to $\sim 1\%$ water over the week of running — directly observable in the data. Water content sets the velocity scale.
- **Air (oxygen) eats the signal.** The measured signal decays exponentially with drift *distance* (attachment length $\lambda \approx 15$ mm at nominal field, consistent with $\sim 1\%$ air), so electrons from the top of the 30 mm gap partially disappear: the recorded charge column is effectively ~ 23 mm. This costs amplitude, not efficiency, at the June contamination level — but it is the reason gas tightness and flushing matter.

9 The chamber edge

The drift field is distorted within $\sim 25\text{--}40$ mm of the frame (the fringe region unavoidable in any bounded drift volume). Consequences, all measured: the efficiency turns on from 0 to 96% over the

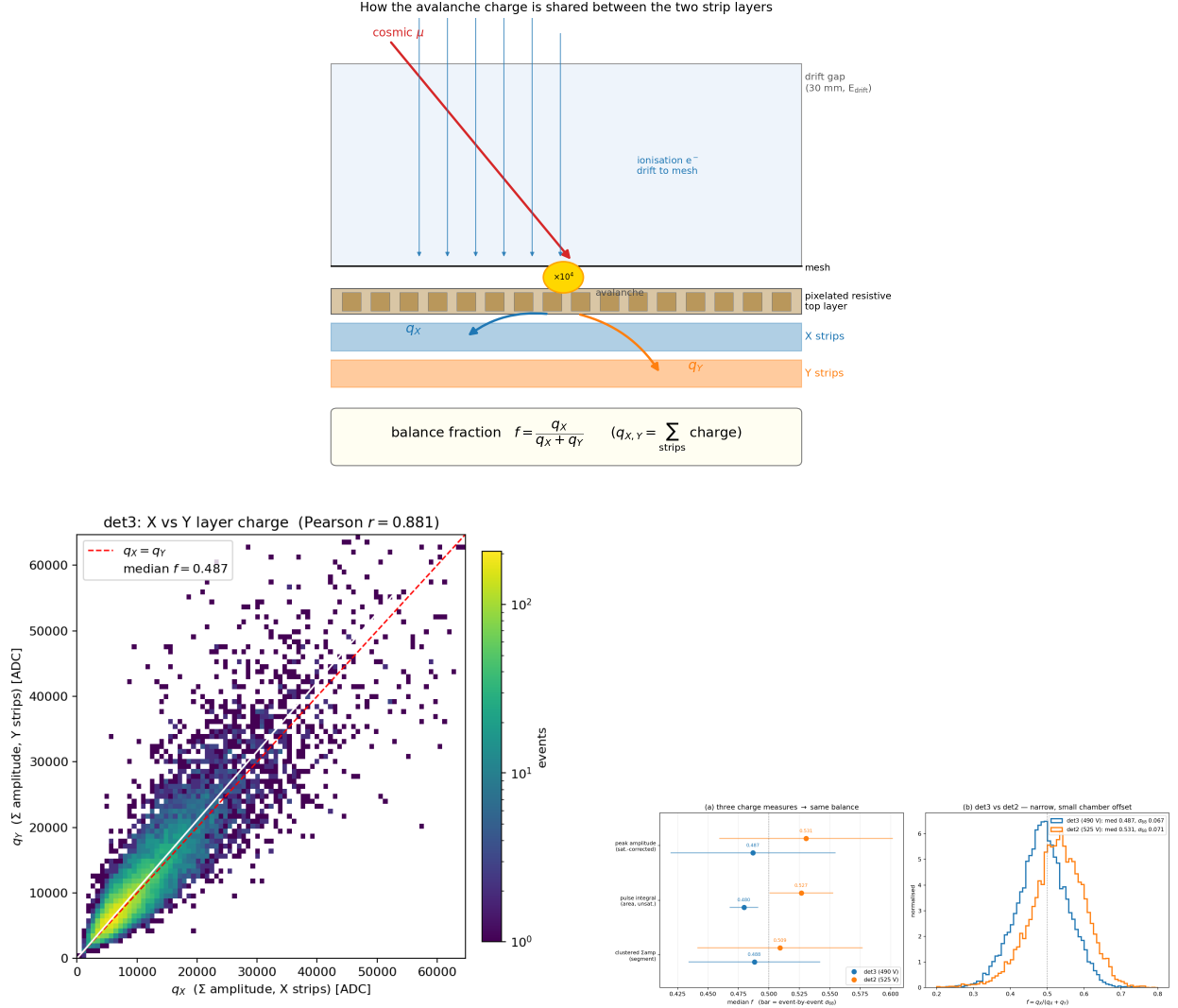


Figure 15: Top: the chamber stack — the pixelated resistive top layer routes the avalanche charge to both strip layers. Bottom left: total X-layer vs. Y-layer charge per event ($r = 0.89$). Bottom right: the balance fraction $f = q_X / (q_X + q_Y)$ is narrow and centred near 0.5 in both chambers measured.

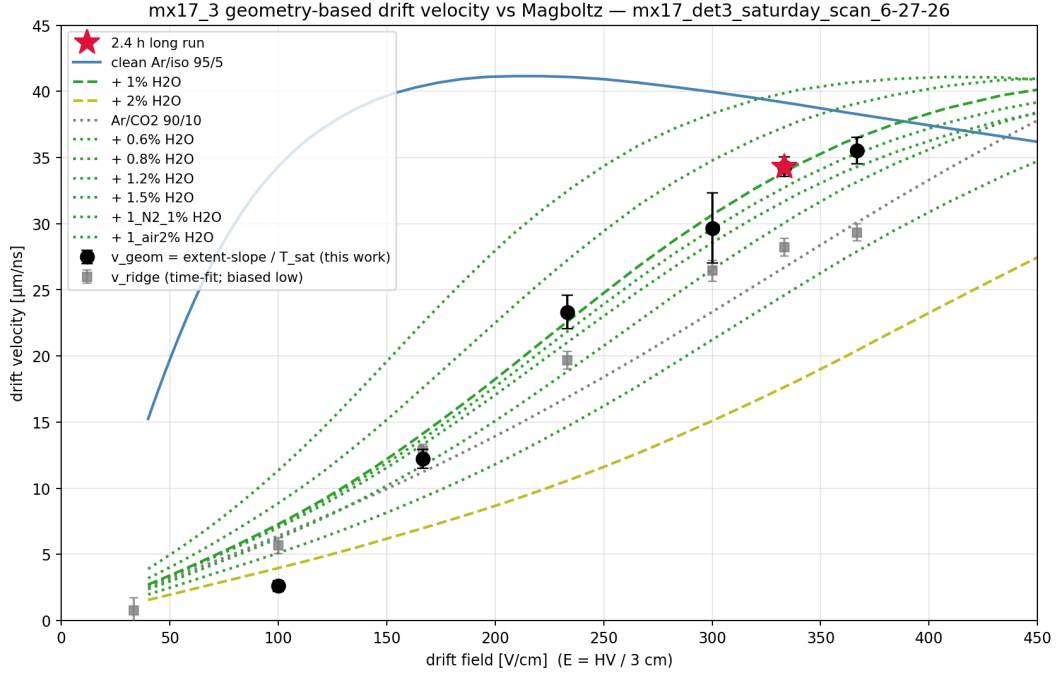


Figure 16: Measured drift velocity vs. drift field, compared with Magboltz simulation for different gas admixtures: the data identify Ar/iso 95/5 with $\sim 1\%$ water.

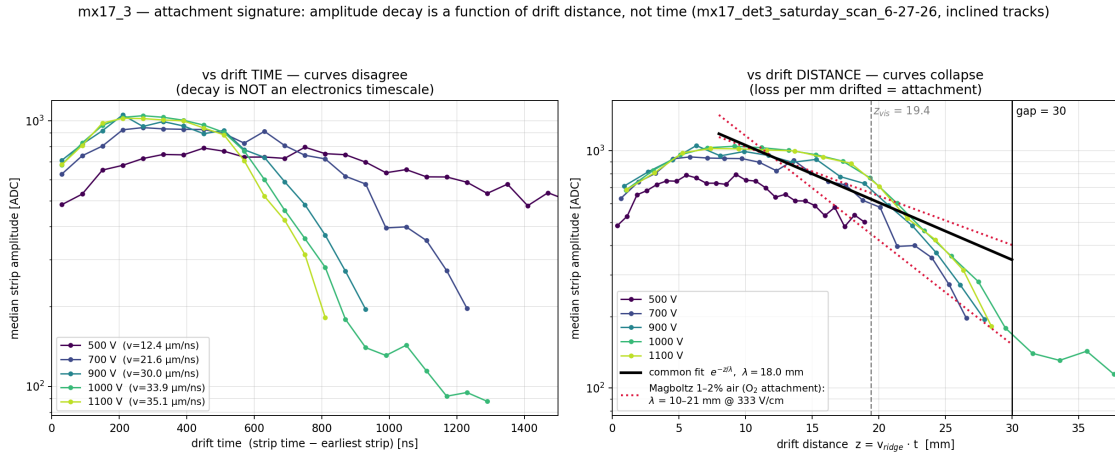


Figure 17: Signal amplitude vs. drift *time* (left; curves at different drift voltages disagree) and vs. drift *distance* (right; all voltages collapse onto one exponential, $\lambda \approx 18$ mm): the loss is per millimetre drifted — electron attachment on oxygen from $\sim 1\%$ air — and it shortens the usable charge column.

first 25 mm from the edge; track angles acquire a $\sim 3^\circ$ inward tilt in the band (position measurements remain good — only angles are affected); and discharges preferentially seed there. Everything inboard of the band is uniform. For the experiment this is handled by a simple fiducial cut.

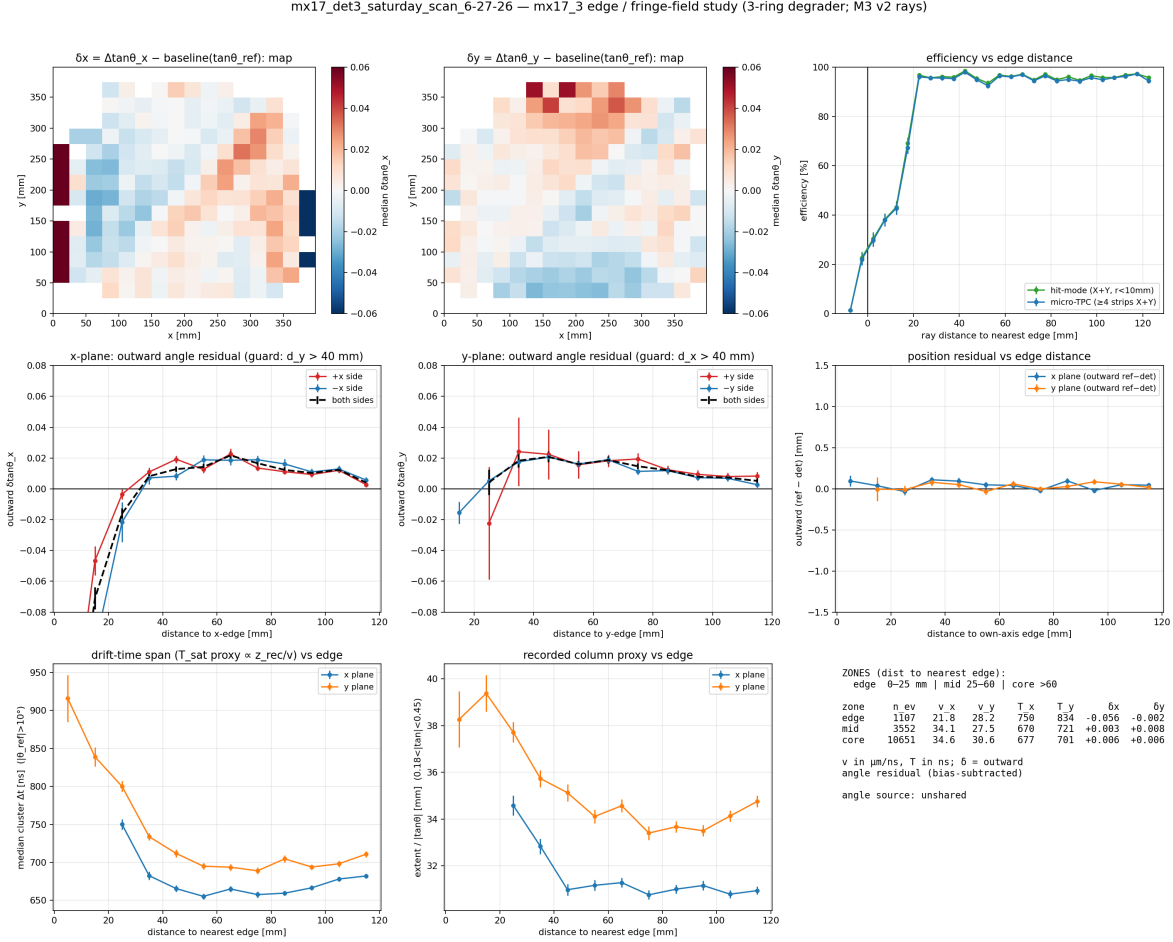


Figure 18: The full edge study (Detector A). The key panel is the efficiency vs. distance-to-edge curve (top right): a 25 mm turn-on band, then a flat $\sim 96\%$ core. Companion panels show the angle tilt and the slowed drift in the same band, and that position residuals stay flat to the edge.

10 The full fleet: five detectors compared

All five chambers were characterized identically. Detector letters are the experiment labels (A–E); the mx17 numbers are the construction labels.

Fleet summary (June 2026, best runs, identical definitions)

Detector	Efficiency	Position σ	Angle σ_{68}	Spark	Verdict
A (mx17_3)	92.9 %	0.47 mm	1.7°	3.9 %	Best performer
B (mx17_2)	91.3 %	0.46 mm	2.3°	5.0 %	Healthy
C (mx17_6)	57.8 %	0.45 mm	3.9° [†]	23.0 %	Spark-limited
D (mx17_7)	43.1 %	0.59 mm	3.4°	33.4 %	Spark-limited
E (mx17_4)	20.7 %	0.67 mm	2.6°	3.2 %	Gain-limited

Efficiency within 5 mm of the reference track, full active area, 2026-07-14 recipe ($\chi^2 < 1.0$, $N_{\text{clus}} = 4$; was $\chi^2 < 5$, $N_{\text{clus}} \geq 3$ – superseded values: A 88.8%/0.63 mm/1.75°, B 87.0%/0.64 mm/2.47°, C 55.7%/0.59 mm/4.14°, D 36.4%/0.86 mm/3.64°, E 20.0%/0.89 mm/2.70°, do not quote). Position σ includes the reference telescope. Angle σ_{68} = the hybrid (current-best) tracking, $|\theta| < 5^\circ$ band, self-trained, odd-eid holdout, run **fleet-wide** (coverage A 98 %, B 96 %, C 98 %, D 90 %, E 79 %). The 700 V spark-limited C/D and gain-limited E are degraded but measured. [†]C is low-angle-only: its X-plane develops no drift-time structure (board-C mesh defect), so the time-fit plateau is unusable and only the head-on signature-regression band is quoted. Spark = % of crossings coinciding with a discharge.

The right way to read this table:

- **A and B are excellent** and consistent with each other — high uniform efficiency at sub-pitch resolution. Where both were probed deeply (charge sharing, charge balance, timing, tracking), they agree, which is evidence the performance is a property of the design.
- **C and D reconstruct beautifully when they fire** (note their position resolutions — C’s 0.45 mm is the best core in the fleet) but were operated with a spark rate of 23–33% of crossings during the June tests; their HV scans show classic turn-on/plateau/falloff with optima at 480 V and 440 V respectively. Their low headline efficiencies are an operating-point effect (drift 700 V, amplification at/past the sparking edge), not broken chambers.
- **E fires on ~70% of muons but reconstructs only ~21%:** its clusters are too small (too little gas gain) to pass reconstruction. This is a gain problem (HV/threshold/gas), the same pathology another chamber showed earlier and recoverable in principle — not a dead detector.

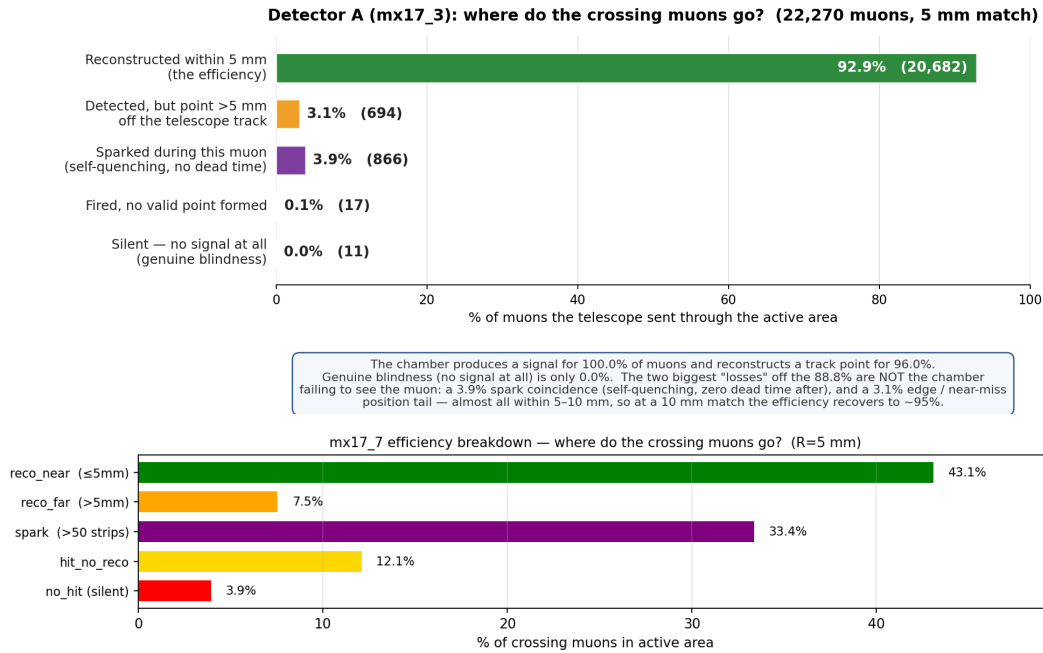


Figure 19: The fleet story in two loss budgets: Detector A (top) loses almost nothing; Detector D (bottom) fires on the muons just as reliably but loses a third of its crossings to discharges at its June operating point. Per-detector efficiency maps for all five chambers are in [figures/ \(2x series\)](#) and the per-detector QA pages in [source_reports/june-detectors-overview.pdf](#).

11 Quotable numbers, with the honest caveats

Quantity	Value	Fine print
Active area	$\sim 40 \times 40 \text{ cm}^2$	512 strips $\times$ 0.78 mm, per coordinate
Strip pitch	0.78 mm	both coordinates
Drift gap	30 mm	operated at $\approx 330 \text{ V/cm}$
Gas	Ar/iC ₄ H ₁₀ 95/5	atmospheric, flushed
Drift velocity (nominal)	$34 \pm 1.5 \mu\text{m/ns}$	measured; matches Magboltz w/ $\sim 1\%$ H ₂ O
Efficiency (A, active area)	$92.9 \pm 0.2 \%$	within 5 mm, operating (spark-free 96.6 %)
Efficiency (A, core)	$\sim 96 \%$	>25 mm from frame (pre-2026-07-14, not yet re-verified)
Muons producing any signal (A)	100.0 %	chamber essentially never blind (silent $\approx 0 \%$)
Position resolution (A)	0.44–0.47 mm	sub-pitch; includes telescope error (conservative)
Angular resolution (A)	$\approx 1.7^\circ$	hybrid tracking, per plane, all angles; bias $\approx 0.1^\circ$
Time resolution (A)	33 ns ($\approx 1 \text{ mm drift}$)	telescope-free, inter-plane method
Absolute event time (A)	$\sigma_{68} = 37.7 \text{ ns}$	vs. scintillator trigger
X/Y charge balance	0.49–0.53, flat	pixel layer routes charge evenly; $r \approx 0.84\text{--}0.89$
Charge sharing (design)	45–52% to neighbour	measured on A and B; enables sub-pitch
Spark rate (A, at optimum)	3.9 % of crossings	Poisson 0.33 Hz; edge-seeded; muon-induced
Dead time after spark	none measurable	efficiency & gain flat vs. time since spark
Optimal amplification HV	440–495 V	per chamber; plateaus $\sim 93\%$ (A/B)

Caveats to keep the talk honest.

1. All results are **preliminary** (June 2026 bench campaign; analysis mature but pre-publication).
2. Position resolutions are **detector** $\oplus$ **telescope**: the reference tracker’s pointing error is not yet deconvolved, so quoted values are upper limits (the planned deconvolution will only improve them).
3. Results are for **cosmic muons** (minimum-ionizing, broad angular spread) at the specific gas and HV listed; the experiment’s particles and settings differ in detail.
4. C, D and E numbers reflect their June *operating points*; their HV scans indicate better performance is available at optimum settings.
5. Angular resolutions are quoted **only from the current-best (hybrid) tracking**, run **fleet-wide** (A 1.7° , B 2.3° , C 3.9° , D 3.4° , E 2.6° ; $|\theta| < 5^\circ$ band). The 700 V chambers C/D and gain-limited E are degraded but honestly measured. **C’s plateau time-fit is unusable** — its X-plane (mx17_6, FEU 3) develops no micro-TPC drift-time structure (a board-C mesh defect), so only C’s low-angle signature-regression band is quoted; the plateau is omitted, not averaged in.

A Where each number comes from

Every value in this document traces to the June-2026 QA analysis (`mx_june_cosmic_qa/` in the analysis repository): `JUNE_RESULTS_SUMMARY.md` (fleet efficiencies, resolutions, HV scans), `PAPER_STATUS.md` and `MICROTPC_RUNBOOK.md` (micro-TPC chain: drift velocity, sharing, angles, position estimators, edge, gas), the standalone reports in `source_reports/` (time resolution, X/Y charge balance, spark studies), and `DET3_WEEKEND_ANALYSIS.md` (detector A deep-dive). The reference tracking is the v2 reprocessing (cleaner track selection), most recently re-tightened 2026-07-14 to $\chi^2 < 1.0$, $N_{\text{clus}} = 4$ (was $\chi^2 < 5$, $N_{\text{clus}} \geq 3$) after finding the position residual was reference-limited, not chamber-limited — see `det3_recofar_analysis/M3_CUT_AND_ACTIVE_AREA_NOTE.md`; all five detectors were re-analyzed with each recipe in turn.

B Mini-glossary

Micromegas Gaseous detector with a micro-mesh separating a drift region from a thin amplification gap.

Resistive strips A resistive layer over the readout that limits discharge currents — the spark-protection scheme validated in Section 6.

micro-TPC mode Using electron drift time as a depth coordinate, so a single chamber measures a 3D track segment, not just a 2D point.

σ_{68} Half-width containing 68% of a distribution — like a Gaussian σ but robust to tails.

Efficiency plateau HV range where efficiency is maximal and flat; the natural operating point.

Attachment Capture of drifting electrons by electronegative contaminants (O_2); shortens the visible charge column.